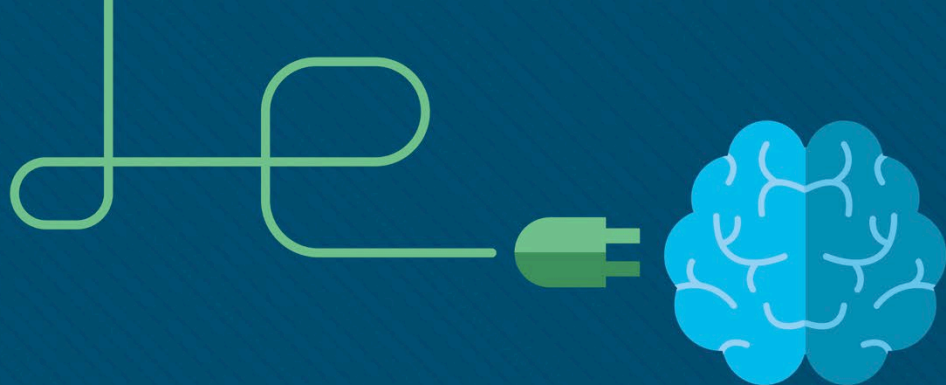




Module 7: DHCPv4

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: DHCPv4

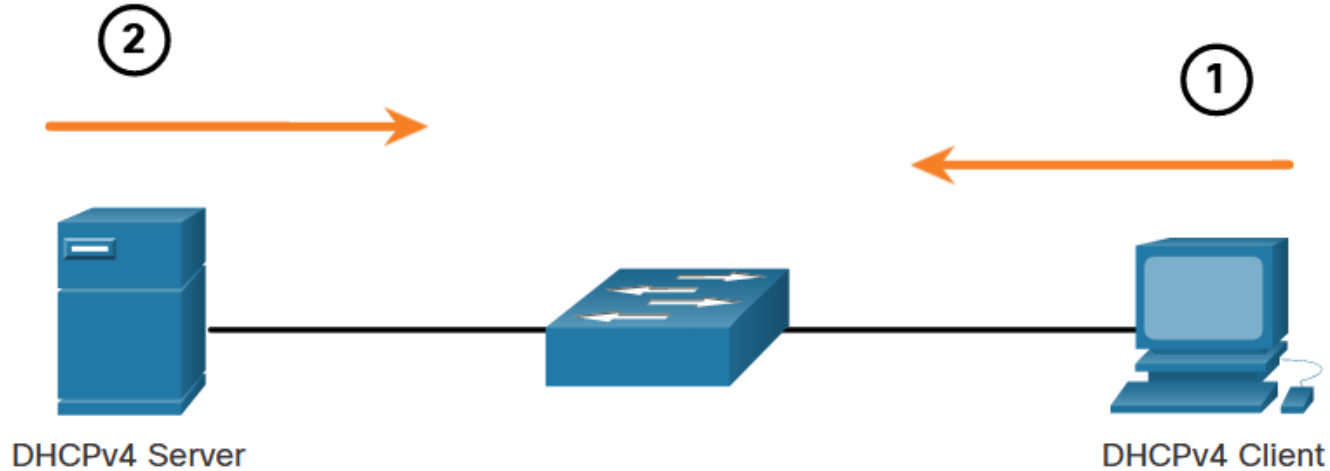
Module Objective: Implement DHCPv4 to operate across multiple LANs

Topic Title	Topic Objective
DHCP4 Concepts	Explain how DHCPv4 operates in a small- to medium-sized business network.
Configure a Cisco IOS DHCP4 Server	Configure a router as a DHCPv4 server.
Configure a DHCP4 Client	Configure a router as a DHCPv4 client.

7.1 DHCPv4 Concepts

DHCPv4 Concepts

DHCPv4 Server and Client



1. The DHCPv4 lease process begins with the client sending a message requesting the services of a DHCP server.
2. If there is a DHCPv4 server that receives the message, it will respond with an IPv4 address and possible other network configuration information.

DHCPv4 Concepts

DHCPv4 Operation

DHCPv4 works in a client/server mode.

When a client communicates with a DHCPv4 server, the server assigns or leases an IPv4 address to that client.

The client connects to the network with that leased IPv4 address until the lease expires.

The client must contact the DHCP server periodically to extend the lease.

This lease mechanism ensures that clients that move or power off do not keep addresses that they no longer need.

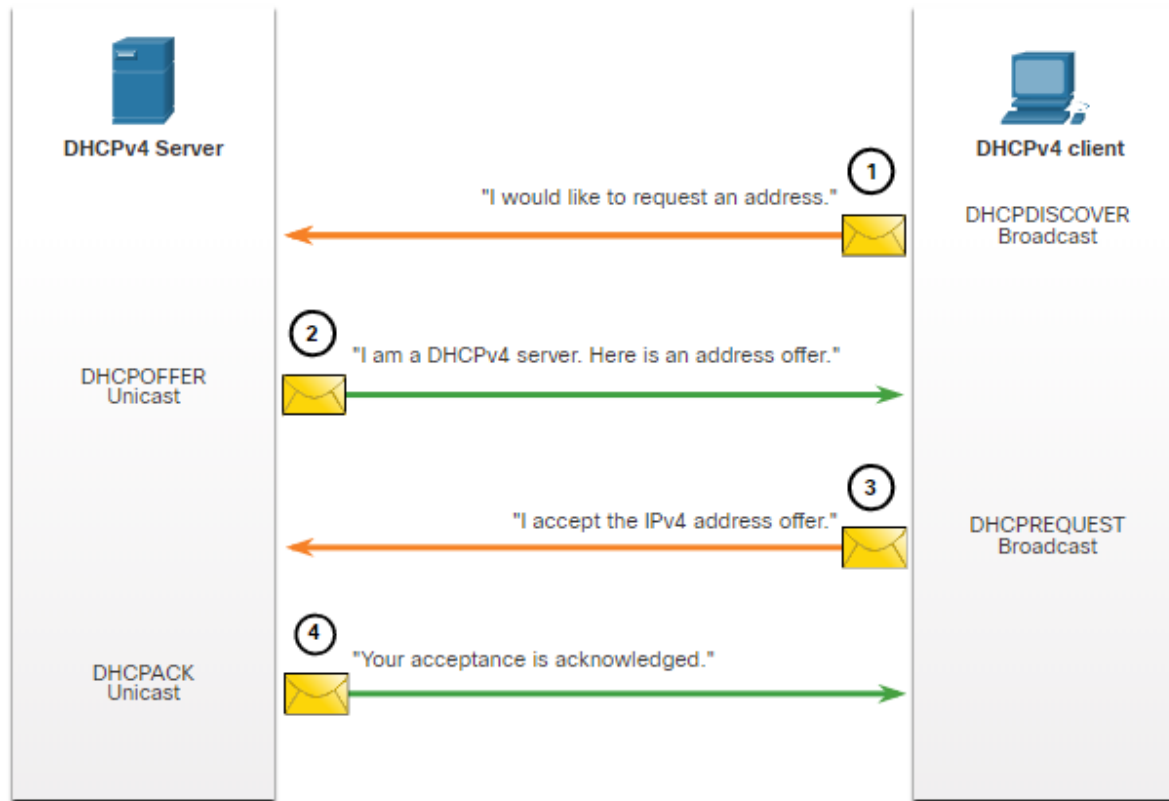
When a lease expires, the DHCP server returns the address to the pool where it can be reallocated as necessary.

DHCPv4 Concepts

Steps to Obtain a Lease

When the client boots (or otherwise wants to join a network), it begins a four-step process to obtain a lease:

1. DHCP Discover (DHCPDISCOVER)
2. DHCP Offer (DHCPOFFER)
3. DHCP Request (DHCPREQUEST)
4. DHCP Acknowledgment (DHCPACK)



DHCPv4 Concepts

Steps to Renew a Lease

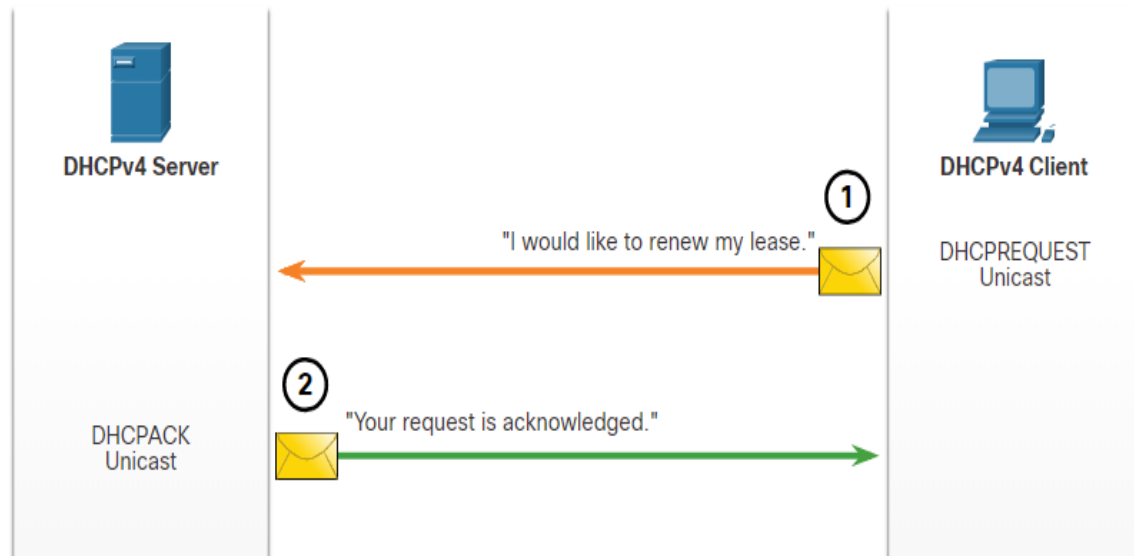
Prior to lease expiration, the client begins a two-step process to renew the lease with the DHCPv4 server, as shown in the figure:

1. DHCP Request (DHCPREQUEST)

Before the lease expires, the client sends a DHCPREQUEST message directly to the DHCPv4 server that originally offered the IPv4 address.

2. DHCP Acknowledgment (DHCPACK)

On receiving the DHCPREQUEST message, the server verifies the lease information by returning a DHCPACK.



Note: These messages (primarily the DHCPOFFER and DHCPACK) can be sent as unicast or broadcast according to IETF RFC 2131.